

PageRank – K-POP Groups and Agencies

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Introduction

- K-Pop's Global Spread
- The Success of Idol Groups Under Major Agencies
- HYBE, SM, JYP, YG
- The Agency with the Greatest Influence on Idol Groups
- The Idol with the Greatest Influence on Their Agency
- What Influence Do They Have Each Other?

Graph Data PageRank: 4-Steps Process



01.

Data Collection
and Cleaning



02.

Graph Modeling



03.

Application of
the PageRank
Algorithm



04.

Result Analysis

[118]:

	Group	Company
0	7 O'clock	Jungle
1	Limitless	ONO
2	VAV	A team
3	Hash Tag	LUK
4	MOMOLAND	Double Kick
...	...	...
1211	SNUPER	Widmay
1293	UNIQ	Yuehua
1308	G-reyish	Elijah
1373	Gfriend	Source
1384	Luminous	Barunson

288 rows × 2 columns

[154]:

Group	status	paks	music_show_awards	end_year_awards	physical_sales	organic_youtube_views
4Minute	Inactive	0	23	7	205088	818287184
AOA	Inactive	0	10	3	223475	872310873
B.A.P	Inactive	0	8	4	911467	301681895
BIGBANG	Inactive	7	84	46	3514210	4589887632
Beast	Inactive	0	47	28	1087745	657013261
Block B	Inactive	0	8	1	418220	353939253
Busker Busker	Inactive	3	10	10	344002	44241494
CLC	Inactive	0	2	0	65774	420286429
Crayon Pop	Inactive	0	2	3	36845	188584801
GFriend	Inactive	1	71	13	999591	1145370316
Girl's Day	Inactive	0	12	4	127226	553105564
H.O.T	Inactive	0	0	0	0	16197860
I.O.I	Inactive	1	9	3	276039	297420417
IZ*ONE	Inactive	0	26	7	2101206	1051316370
LOONA	Inactive	0	4	1	698360	365528056
Miss A	Inactive	2	16	12	111748	502828620
Momoland	Inactive	0	11	4	66422	1881942223
NU'EST	Inactive	0	17	5	1176812	439100514
NewJeans	Inactive	2	35	32	8278782	2296351136
PENTAGON	Inactive	0	2	0	650777	733366148
SS501	Inactive	0	12	5	398819	225868285
Sistar	Inactive	2	48	25	137519	957775756
Trouble Maker	Inactive	0	12	2	66974	248786656
VICTON	Inactive	0	3	0	750171	180700492
WJSN	Inactive	0	9	1	850460	241820625
Wanna One	Inactive	0	49	21	3687191	578452979
Weeekly	Inactive	0	0	1	314973	318429943
Wonder Girls	Inactive	1	32	17	309310	586428007
f(x)	Inactive	0	33	10	548777	560004408

Name: count, dtype: int64

Data Collection and Cleaning

- Collect only necessary data
- Cleaning Up Missing Data
- Check for consistency with data information

Group	Company	followers	top_track_popularity	status	paks	music_show_awards	end_year_awards	physical_sales	Youtube_view_count
f(x)	SM	0.000000	0.000000	Inactive	0.0	0.018043	0.010515	0.001798	0.004458
EXO	SM	0.039178	0.036308	Active	0.0	0.062329	0.077813	0.044772	0.034409
NCT	SM	0.027920	0.028871	Active	0.0	0.021870	0.016824	0.046155	0.014526
Super Junior	SM	0.000000	0.000000	Active	0.0	0.028978	0.051525	0.020814	0.015861
aespa	SM	0.013805	0.036745	Active	2.0	0.019683	0.042061	0.028476	0.020428
Super Junior-M	SM	0.000000	0.000000	0	0.0	0.000000	0.000000	0.000000	0.000000
Red Velvet	SM	0.034624	0.034121	Active	1.0	0.046473	0.019979	0.017242	0.027034
SHINee	SM	0.012277	0.033246	Active	0.0	0.037726	0.035752	0.010240	0.010418

Graph Modeling

- Modeling the organized data as a graph
- nodes: Idol Groups, Agencies
- Weights: other columns
- Determining Appropriate Weights

	node	rank
0	SM	0.031592
1	HYBE	0.023512
2	YG	0.018140
3	JYP	0.018005
4	Starship	0.007357
5	FNC	0.007357
6	KQ	0.005936

	node	rank
0	SM	0.030341
1	JYP	0.029680
2	YG	0.025728
3	HYBE	0.022580
4	Starship	0.007065
	FNC	0.007065
	node	rank
	JYP	0.025630
1	HYBE	0.022837
2	SM	0.022837
3	YG	0.020044
4	Starship	0.014458
5	FNC	0.014458
6	BTS	0.014342
7	BLACKPINK	0.013231

```
import networkx as nx

G = nx.DiGraph()

for i, row in my_data_set_4_dropped.iterrows():
    group = row['Group']
    company = row['Company']

    #weight = row['Youtube_view_count'] * 0.7 + row['revenue'] * 0.3
    G.add_node(group, node_type='group')
    G.add_node(company, node_type='company')

    G.add_edge(group, company, weight=(row['music_show_awards'] * 0.4) + (row['end_year_awards'] * 0.4) + (row['physical_sales'] * 0.2))
    G.add_edge(company, group, weight=(row['followers'] * 0.2) + (row['top_track_popularity'] * 0.2) + (row['Youtube_view_count'] * 1.0))
    #G.add_edge(company, group, weight=(row['Youtube_view_count'] * 1.0) + (row['top_track_popularity'] * 1.0))

#print(G.number_of_nodes())
#print(G.number_of_edges())

pagerank = nx.pagerank(G, alpha=0.85, tol=1e-8, max_iter=1000, weight='weight')

sorted_pagerank = sorted(pagerank.items(), key = lambda x: x[1], reverse=True)
#top_20_pagerank = sorted(pagerank.items(), key = lambda x: x[1], reverse=True)[:20]

top_20_pagerank_1 = sorted(pagerank.items(), key = lambda x: x[1], reverse=True)[:20]

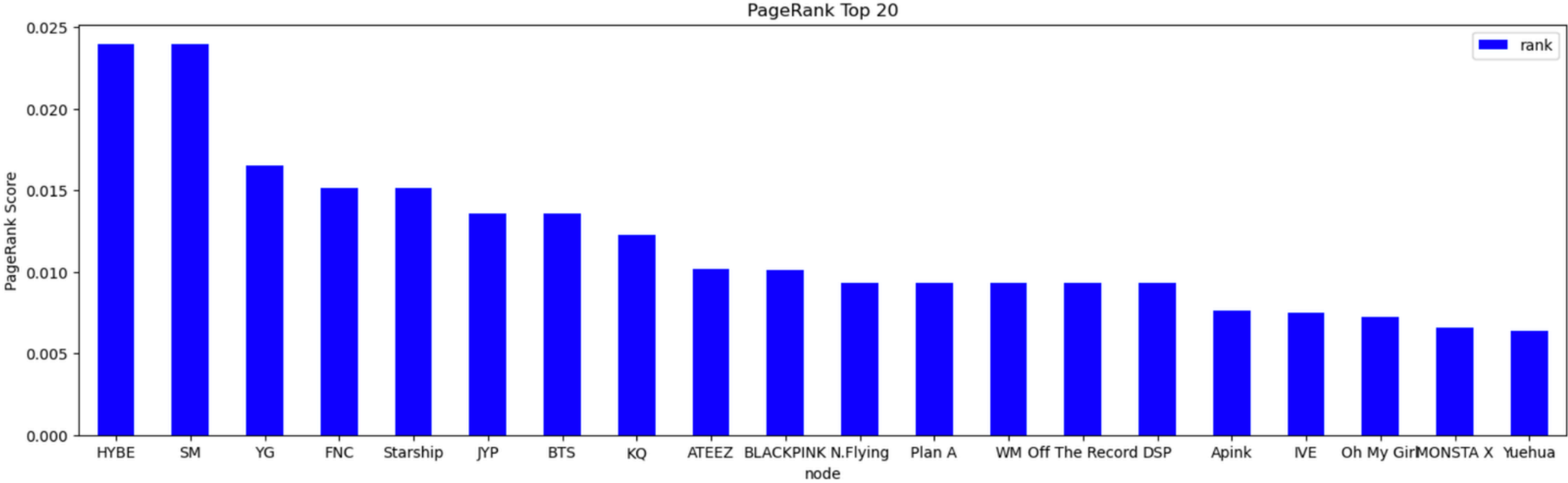
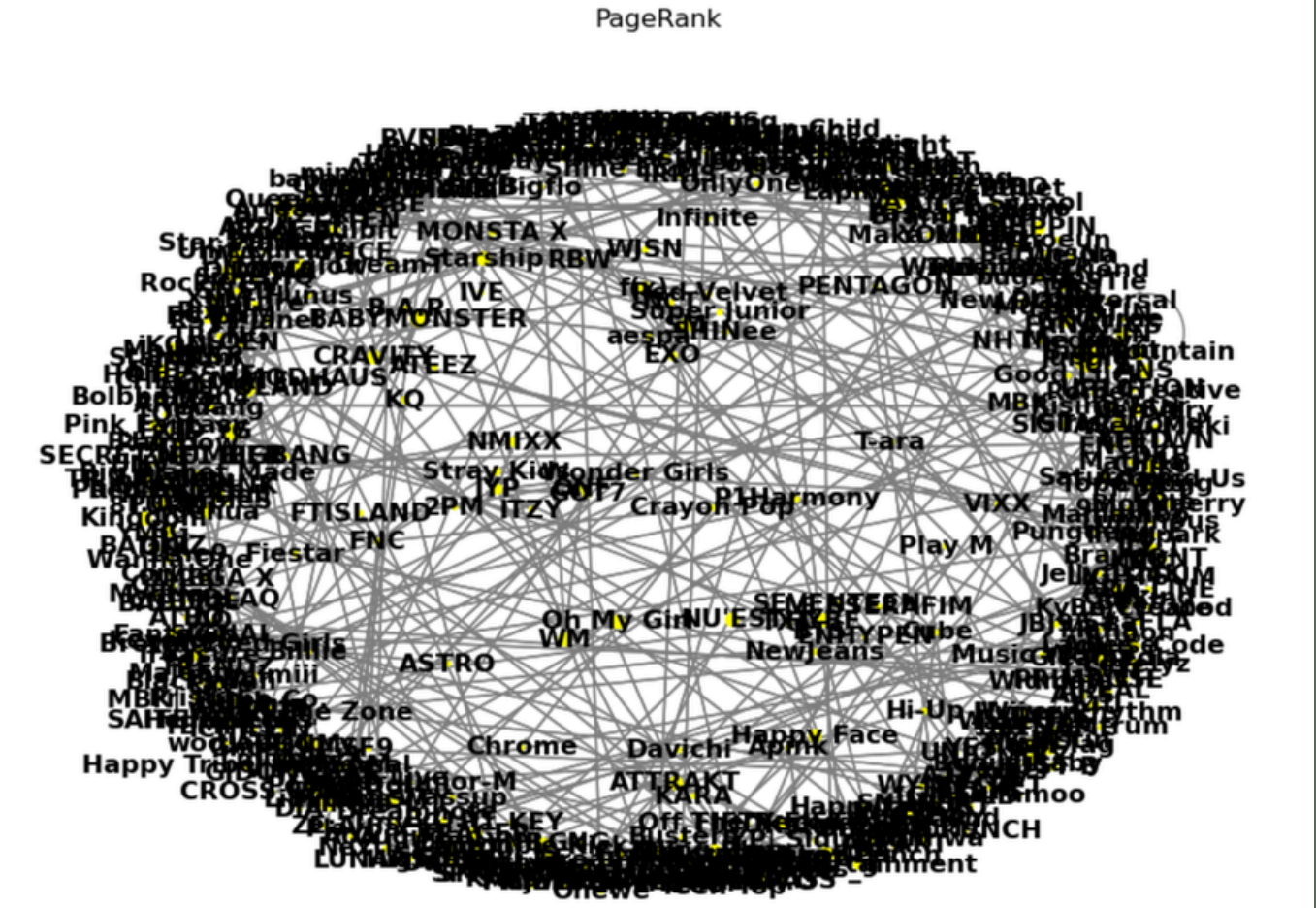
top_20_pagerank_11 = pd.DataFrame(top_20_pagerank_1, columns=['node', 'rank'])
top_20_pagerank_11
```

	Company	followers	top_track_popularity	status	paks	music_show_awards	end_year_awards	physical_sales	Youtube_view_count	revenue
0	SM	0.000000	0.000000	Inactive	0.0	0.018043	0.010515	0.001798	0.004458	0.216
1	SM	0.039178	0.036308	Active	0.0	0.062329	0.077813	0.044772	0.034409	0.216
2	SM	0.027920	0.028871	Active	0.0	0.021870	0.016824	0.046155	0.014526	0.216
3	SM	0.000000	0.000000	Active	0.0	0.028978	0.051525	0.020814	0.015861	0.216
4	SM	0.013805	0.036745	Active	2.0	0.019683	0.042061	0.028476	0.020428	0.216
5	SM	0.000000	0.000000	0	0.0	0.000000	0.000000	0.000000	0.000000	0.216
6	SM	0.034624	0.034121	Active	1.0	0.046473	0.019979	0.017242	0.027034	0.216
7	SM	0.012277	0.033246	Active	0.0	0.037726	0.035752	0.010240	0.010418	0.216

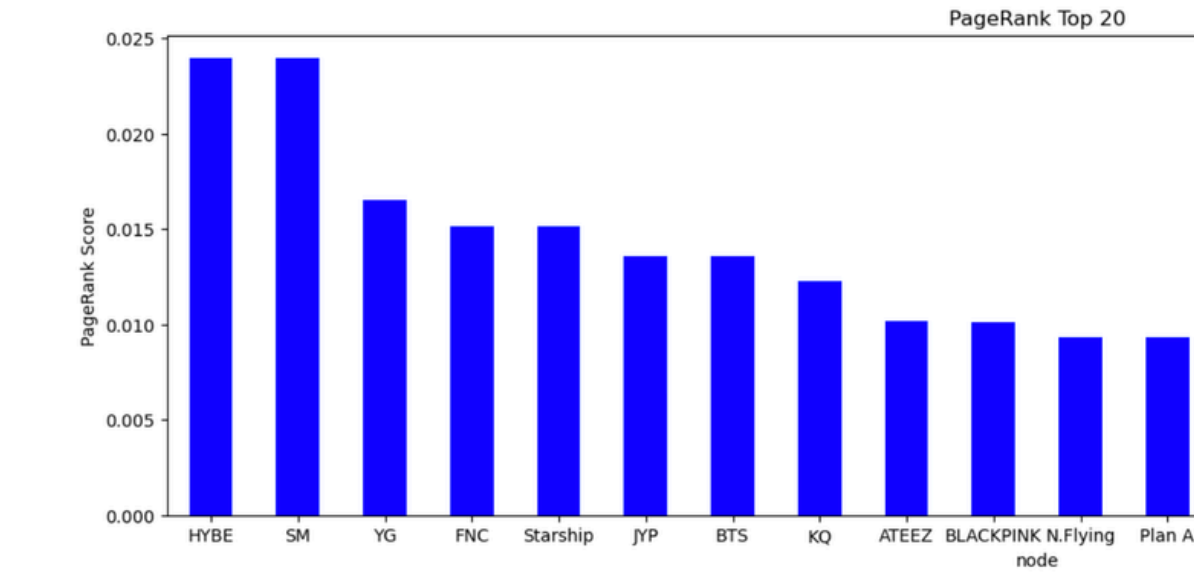
Application of PageRank Algorithm

[183]:

	node	rank
0	HYBE	0.023968
1	SM	0.023968
2	YG	0.016538
3	FNC	0.015174
4	Starship	0.015174
5	JYP	0.013610
6	BTS	0.013551
7	KQ	0.012242
8	ATEEZ	0.010200
9	BLACKPINK	0.010140
10	N.Flying	0.009351
11	WM	0.009311
12	Plan A	0.009311
13	Off The Record	0.009311



Application of PageRank Algorithm



[183]:

	node	rank
0	HYBE	0.023968
1	SM	0.023968
2	YG	0.016538
3	FNC	0.015174
4	Starship	0.015174
5	JYP	0.013610
6	BTS	0.013551
7	KQ	0.012242
8	ATEEZ	0.010200
9	BLACKPINK	0.010140
10	N.Flying	0.009351
11	WM	0.009311
12	Plan A	0.009311
13	Off The Record	0.009311
14	DSP	0.009311
15	Apink	0.007625
16	IVE	0.007521
17	Oh My Girl	0.007234
18	MONSTA X	0.006601
19	Happy Face	0.006380

01

Groups → Agencies
Weights: Music Award * 0.4 + End of Year Award * 0.4 + Physical Sales * 0.2

02

Agencies → Groups
Weights: Followers * 0.2 + Top Track Popularity * 0.2 + Youtube View Counts * 0.6

03

HYBE: 0.023968
SM: 0.023968

BTS: 0.013551

Conclusion

Agencies: HYBE, SM
Idol Group: BTS

References

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Thank You
